

Owen Hyatt

Final Project Report

The subject I chose for this project was a plain, white ceramic mug. I was warned of the difficulty of capturing something like the handle of the mug, and was told that it may make it more difficult to get an accurate 3D model. To an extent, this was true throughout the process.

I initially struggled to use the Meshroom software, as my laptop was seemingly unable to handle the software and I had repeated issues with crashing. I was then informed about two mobile apps that I could use in order to capture and more accurately display my 3D model. The first was PolyCam, which required me to upload my same pictures of the mug (I have attached three of them below for reference) and worked similarly to Meshroom in many ways.

Example photos:





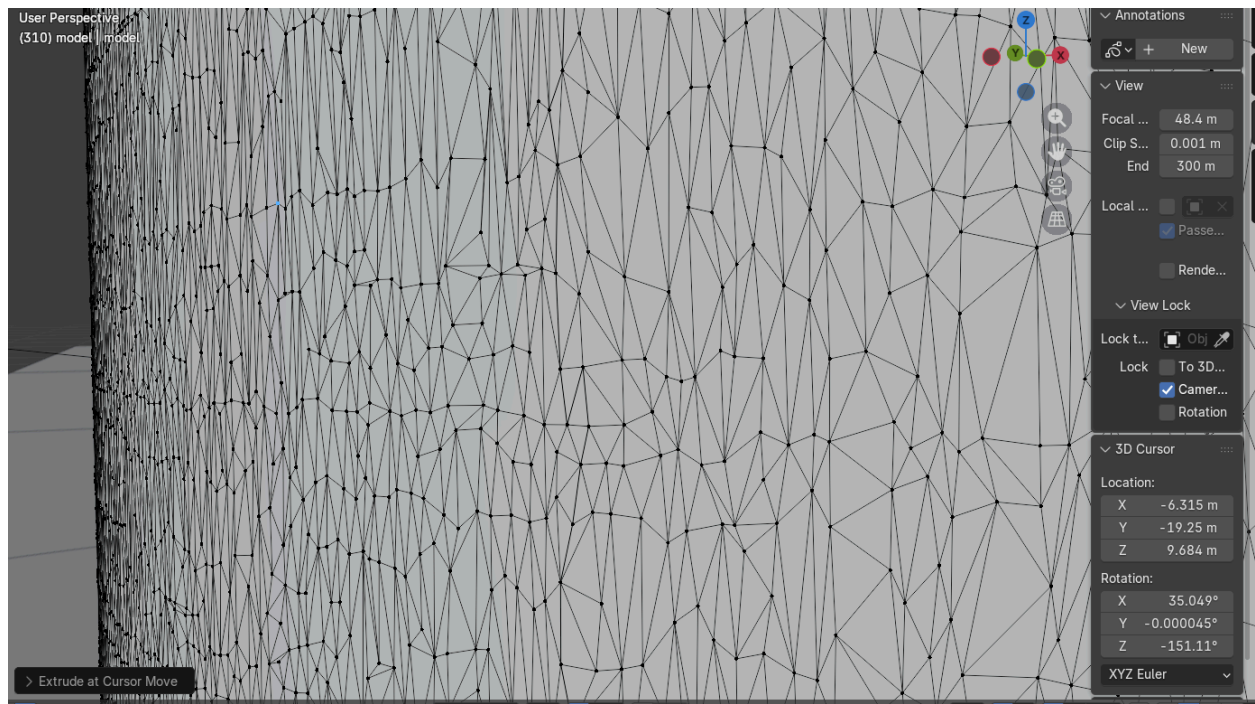


In retrospect, and from speaking to some of my classmates regarding this project, I believe my setting of the photos I took was a large reason for why many of the models I had generated using this app and these photos did not pan out well. It was at this point I realized I had to scrap these photos and come up with a new approach.

This is when I found the mobile app Scaniverse. This app was different, as it allowed scanning through video. I used this app to slowly scan every side of my subject, while also practicing better lighting than before in order to make sure the model turned out as good as it possibly could. The model scanned decently well into the app, which is eventually what I used for my final model, which is shown below.



I still believe I had show issues with lighting, and because I used an OBJ file, this came back to be an issue a bit later. I then uploaded this file to my laptop as an .OBJ file, while maintaining some of the textures of the original scanned object (things I learned through research that were not true for STL or PLY files). The final 3D model was then edited within Blender entirely. By using the Edit mode on the model, I was able to directly edit vertices, smooth surfaces, close any gaps, and clean up any failing details around the 3D model. This is where the differences between the Scaniverse version of the model and the final model come from. Many of the techniques I used were also around the time of Blender Activity 5, which had a focus on editing the shape of 3D models manually, and because I wanted to control the final model and shape it as accurately as I could, I decided to take this manual route in order to create the final model. One of the bigger issues with the initial scan was having a bit of the wooden table on the edges of the bottom of the model. Using the vertex editor in blender, as shown below, it made it simple to drag and manually transform the model. Example of vertices editing:



One issue I had was that the ceramic texture of the mug was seemingly lost in translation. While the mug was still white, it did not look as I expected it to once I had uploaded it into Blender and rendered it. For this reason, I used more I had learned through the Blender Activities and looked online (through the links provided for activity 6) for a new, ceramic texture for the mug. The one I eventually used is attached to the assignment. While this still failed to capture the true ceramic, glossy texture I desired, I still thought this was necessary to mention and a good step in trying to capture the full realism of the model.

These are some of the versions of the mug that looked “fake” to me, where I felt the need to apply the new texture:

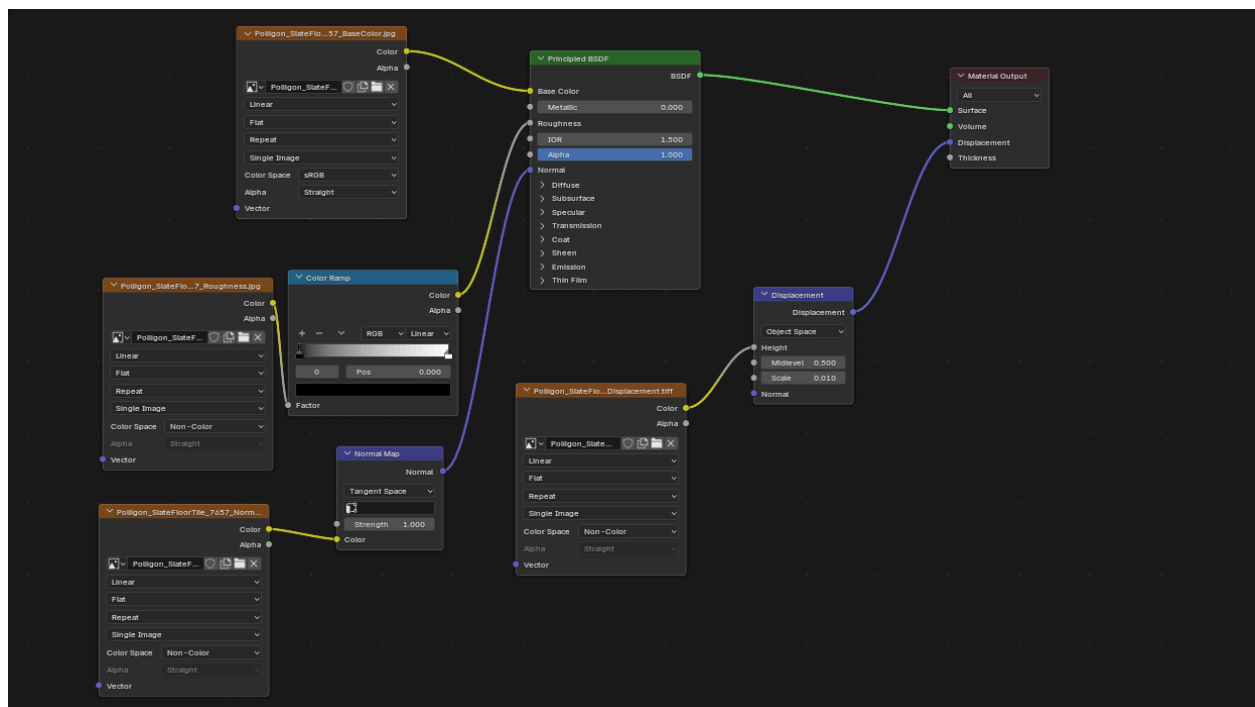




Once the mug itself was complete, I was excited to start working in Blender to create my scene. I started by importing the mug and attempting to decide on what it should do, or interact with. I decided to incorporate a coffee table, which felt like a natural choice given the subject. I found a model for a coffee table online, which is also attached separately to the assignment, and

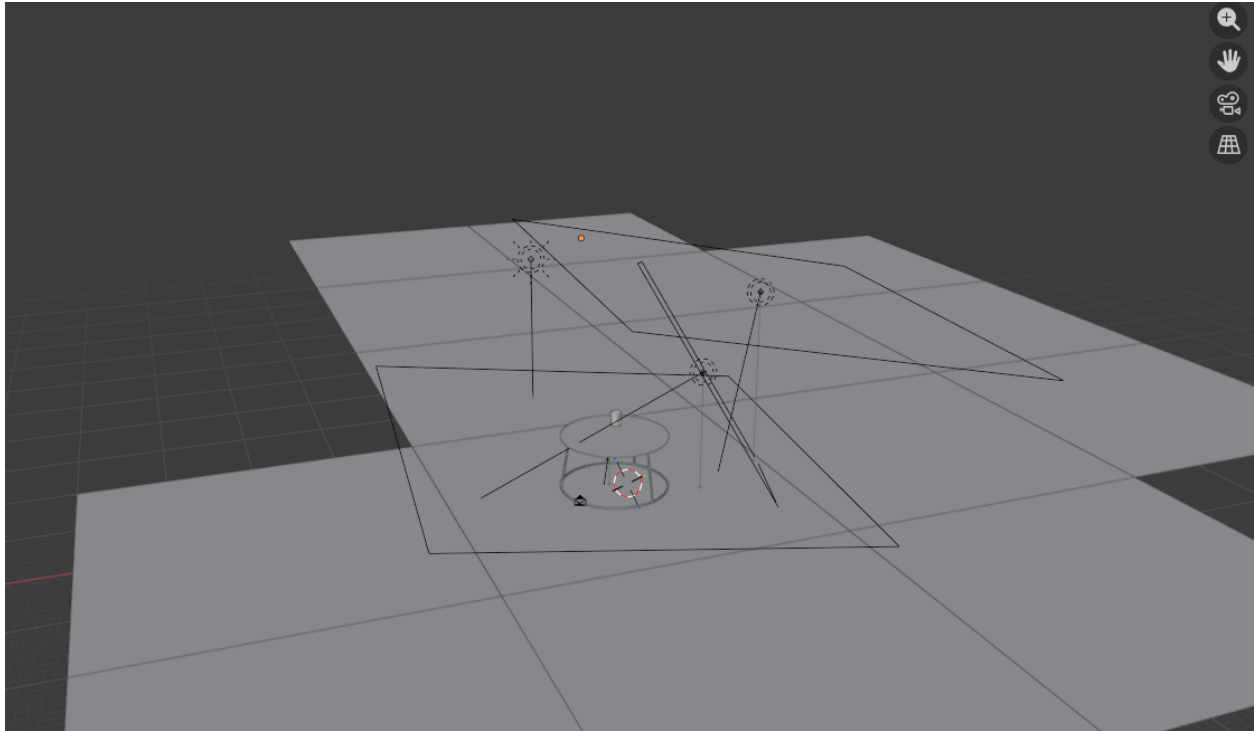
placed this model at the center of the scene, scaling it appropriately and moving the position of the camera to accommodate this larger scene. I then placed the mug on top of the table.

The next challenge came from lighting the scene properly. Initially, I had failed to create any form of floor or plane beneath the table, so when I tried to add a sun or an area light, there was not much more than a very dark scene that was confusing to look at. Once I realized a floor would enhance the scene dramatically, I got to thinking about what I could use to accomplish this, and then decided I wanted to add a tiled floor beneath the table. This was done by inserting a plane mesh and scaling it appropriately, and then looking online for a tiled texture I could use for the floor. The used texture is once again attached to the assignment separately. I also used what I learned from Blender Activity 6 in order to accomplish this. An example screenshot of how I attached the texture is shown below (using base color, roughness, displacement, normal map, color ramp).



Once I had finished the floor, the next step was to animate the scene. I have some brief experience with keyframes in Davinci Resolve in the past. Because of this, I was fairly comfortable with how to do this within Blender itself once I had located the timeline and learned how to apply keyframes themselves. I also had used what I had learned in the Blender activities to know how to scale, move, rotate, and physically mold the object within Blender itself. Once I had done this, the animation process was fairly straightforward. I made sure to rotate the object every way possible, as slowly as possible in most cases, and then slowly lifted it and made it grow in each direction separately. For instance, it stretches itself in the X direction, then the Y direction, and then the Z direction. Finally, I wanted the subject to interact with its environment in some way. This is why, at the end of the video, the subject goes beneath the table, which shows its interaction with both the object itself, and the lighting within the scene.

In the scene, I set up one sun light above the scene, and multiple area lights to simulate being in an outdoor patio area, with both sun light and overhead lights. A screenshot of the final scene itself, with all of these elements, is shown below.



Rendering the project took longer than expected. My final project was over 800 frames, which took several hours in order to complete the render of. Once I had rendered the animation, I took the rendered frames and imported them back into Blender using the video editor. I used the image sequence option in order to combine them into a video (as I initially had not exported as a video and instead exported the animation as its individual frames). This rendered very fast, and while it was a slight challenge, it was a very easy one to resolve.

The final animation is attached to this youtube link: <https://youtu.be/b8gldlQUuac>

This project showed me a lot about computer graphics. My main takeaway is a newfound appreciation I have for animation and 3D modeling; the process is both tenuous and unforgiving, and can be extremely difficult and frustrating at times. However, it can also be extremely rewarding, and expensive. I had the thought, for instance, about how Pixar animates movies while I was animating my scene, which made me laugh but also made me appreciate the work people do in order to bring such complex models and characters to life through animation. Some

of my favorite shows and movies as a child were animated as well; I think this project and process as a whole has given me a new appreciation for them as well, and a new perspective on animation as a whole.

Sources:

Buchholz, Carrie. "How to Animate in Blender: The Ultimate Guide | Skillshare Blog."

Skillshare Blog, 6 Nov. 2023,

[www.skillshare.com/en/blog/how-to-animate-in-blender-the-ultimate-guide/?srltid=Afm](https://www.skillshare.com/en/blog/how-to-animate-in-blender-the-ultimate-guide/?srltid=AfmBOoqijrIrQGSc_INkrVrz-M21-XT-m34eZMJ7O9YUdiwivciRhd9k)

[BOoqijrIrQGSc_INkrVrz-M21-XT-m34eZMJ7O9YUdiwivciRhd9k](https://www.skillshare.com/en/blog/how-to-animate-in-blender-the-ultimate-guide/?srltid=AfmBOoqijrIrQGSc_INkrVrz-M21-XT-m34eZMJ7O9YUdiwivciRhd9k). Accessed 4 May

2026.

MikaelaHolmesFlora & Faun. "Shooting for Photogrammetry." *Instructables*, 2026,

www.instructables.com/Shooting-for-Photogrammetry/.

Poliigon. "Poliigon." *Www.poliigon.com*, www.poliigon.com/.

Stevenson, Kerry. "Hands on with Scaniverse «Fabbaloo." *Fabbaloo*, 2022,

www.fabbaloo.com/news/hands-on-with-scaniverse.

TutsByKai. "Add a Texture to an Object - Blender Tutorial." *Youtu.be*, 2026,

youtu.be/mURA2g1rOSc?si=43E8bZ00KcPdRvgy. Accessed 4 May 2026.

Blender Activities 1-6.

Scaniverse App.

Blender.